

SOLIDWORKS API (.NET VBA C#) — Reference (185/337): IModelDoc2 (part 8 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 8 of 17) (Type)

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GridOptions Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : GridOptions Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SetGridOptions](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub GridOptions( _  
    ByVal DispGrid As System.Boolean, _  
    ByVal GridSpacing As System.Double, _  
    ByVal Snap As System.Boolean, _  
    ByVal DotStyle As System.Boolean, _  
    ByVal NMajor As System.Short, _  
    ByVal NMinor As System.Short, _  
    ByVal Align2edge As System.Boolean, _  
    ByVal AngleSnap As System.Boolean, _  
    ByVal AngleUnit As System.Double, _  
    ByVal MinorAuto As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim DispGrid As System.Boolean  
Dim GridSpacing As System.Double  
Dim Snap As System.Boolean  
Dim DotStyle As System.Boolean  
Dim NMajor As System.Short  
Dim NMinor As System.Short  
Dim Align2edge As System.Boolean  
Dim AngleSnap As System.Boolean  
Dim AngleUnit As System.Double  
Dim MinorAuto As System.Boolean  
  
instance.GridOptions(DispGrid, GridSpacing, Snap, DotStyle, NMajor, NMinor, Align2edge, AngleSnap, AngleUnit, MinorAuto)
```

C#

```
void GridOptions(  
    System.bool DispGrid,  
    System.double GridSpacing,  
    System.bool Snap,  
    System.bool DotStyle,  
    System.short NMajor,  
    System.short NMinor,  
    System.bool Align2edge,  
    System.bool AngleSnap,  
    System.double AngleUnit,  
    System.bool MinorAuto  
)
```

C++/CLI

```
void GridOptions(  
& System.bool DispGrid,  
& System.double GridSpacing,  
& System.bool Snap,  
& System.bool DotStyle,  
& System.short NMajor,  
& System.short NMinor,
```

```
& System.bool Align2Edge,  
& System.bool AngleSnap,  
& System.double AngleUnit,  
& System.bool MinorAuto  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DispGrid

True to display the grid, false to not

GridSpacing

Snap distance

Snap

True to snap to grid, false to not

DotStyle

True for dotted grids, false for not

NMajor

Number of minors in major

NMinor

Number of snaps in minor

Align2Edge

True if to align to an edge, false to not

AngleSnap

True to snap to angle, false to not

AngleUnit

Value of angle to which to snap

MinorAuto

True if the minor grids are to be set automatically, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::GridOptions](#).

Remarks

The Align2Edge argument aligns the grid with the currently selected edge. If Align2Edge is set to True, then you must have selected an edge.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetGridSettings Method](#)
[IModelDoc2::ToolsGrid Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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HideComponent2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideComponent2 Method (IModelDoc2)

Hides the selected component.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideComponent2()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
```



```
instance.HideComponent2()
```

C#

```
void HideComponent2()
```

C++/CLI

```
void HideComponent2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideComponent2](#).

Example

[Undo Hidden Component and Fire Post-Notify Event \(VB60\)](#)
[Undo Hidden Component and Fire Post-Notify Event \(VB.NET\)](#)
[Undo Hidden Component and Fire Post-Notify Event \(C#\)](#)

Remarks

This method is only available for [IDrawingDoc](#) and [IAssemblyDoc](#) objects; it is not available for [IPartDoc](#) objects.
To show a selected component, call [IModelDoc2::ShowComponent2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ShowComponent2 Method](#)
[IModelDoc2::IsActive Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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HideCosmeticThread Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideCosmeticThread Method (IModelDoc2)

Hides the selected cosmetic thread.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideCosmeticThread()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.HideCosmeticThread()
```

C#

```
void HideCosmeticThread()
```

C++/CLI

```
void HideCosmeticThread();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideCosmeticThread](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ShowCosmeticThread Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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HideDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideDimension Method (IModelDoc2)

Hides the selected dimension in this document.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideDimension()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.HideDimension()
```

C#

```
void HideDimension()
```

C++/CLI

```
void HideDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideDimension](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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HideFeatureDimensions Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideFeatureDimensions Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::GetUserPreferenceToggle](#) or [IModelDoc2::SetUserPreferenceToggle](#) and [swDisplayFeatureDimensions](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub HideFeatureDimensions()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.HideFeatureDimensions()
```

C#

```
void HideFeatureDimensions()
```

C++/CLI

```
void HideFeatureDimensions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideFeatureDimensions](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

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HideShowBodies Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideShowBodies Method (IModelDoc2)

Sets whether to hide or show the bodies in the model.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideShowBodies()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.HideShowBodies()
```

C#

```
void HideShowBodies()
```

C++/CLI

```
void HideShowBodies();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideShowBodies](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::HideSolidBody Method](#)
[IFeatureManager::HideBodies Method](#)

[IModelDoc2::ShowSolidBody Method](#)
[IBody2::HideBody Method](#)
[IBody2::DisableDisplay Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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HideSolidBody Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HideSolidBody Method (IModelDoc2)

Hides the currently selected solid body.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideSolidBody()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.HideSolidBody()
```

C#

```
void HideSolidBody()
```

C++/CLI

```
void HideSolidBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HideSolidBody](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::HideShowBodies Method](#)
[IModelDoc2::ShowSolidBody Method](#)
[IFeatureManager::HideBodies Method](#)
[IBody2::HideBody Method](#)
[IBody2::DisableDisplay Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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HoleWizard Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : HoleWizard Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::HoleWizard2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub HoleWizard( _  
    ByVal Depth As System.Double, _  
    ByVal EndType As System.Short, _
```

```

ByVal Flip As System.Boolean, _
ByVal Dir As System.Boolean, _
ByVal HType As System.Integer, _
ByVal D1 As System.Double, _
ByVal D2 As System.Double, _
ByVal D3 As System.Double, _
ByVal D4 As System.Double, _
ByVal D5 As System.Double, _
ByVal D6 As System.Double, _
ByVal D7 As System.Double, _
ByVal D8 As System.Double, _
ByVal D9 As System.Double, _
ByVal D10 As System.Double, _
ByVal D11 As System.Double, _
ByVal D12 As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Depth As System.Double
Dim EndType As System.Short
Dim Flip As System.Boolean
Dim Dir As System.Boolean
Dim HType As System.Integer
Dim D1 As System.Double
Dim D2 As System.Double
Dim D3 As System.Double
Dim D4 As System.Double
Dim D5 As System.Double
Dim D6 As System.Double
Dim D7 As System.Double
Dim D8 As System.Double
Dim D9 As System.Double
Dim D10 As System.Double
Dim D11 As System.Double
Dim D12 As System.Double

```

```

instance.HoleWizard(Depth, EndType, Flip, Dir, HType, D1, D2, D3, D4, D5, D6, D7, D8, D9, D10,
D11, D12)

```

C#

```

void HoleWizard(
    System.double Depth,
    System.short EndType,
    System.bool Flip,
    System.bool Dir,
    System.int HType,
    System.double D1,
    System.double D2,
    System.double D3,
    System.double D4,
    System.double D5,
    System.double D6,
    System.double D7,
    System.double D8,
    System.double D9,
    System.double D10,
    System.double D11,
    System.double D12
)

```

C++/CLI

```

void HoleWizard(
    & System.double Depth,
    & System.short EndType,
    & System.bool Flip,
    & System.bool Dir,

```

```
& System.int HType,  
& System.double D1,  
& System.double D2,  
& System.double D3,  
& System.double D4,  
& System.double D5,  
& System.double D6,  
& System.double D7,  
& System.double D8,  
& System.double D9,  
& System.double D10,  
& System.double D11,  
& System.double D12  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Depth

EndType

Flip

Dir

HType

D1

D2

D3

D4

D5

D6

D7

D8

D9

D10

D11

D12

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::HoleWizard](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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IActiveView Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IActiveView Property (IModelDoc2)

Gets the current active model view in read-only mode.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IActiveView As ModelView

Visual Basic (Usage)

 Copy Code

```
Dim instance As IModelDoc2
Dim value As ModelView

value = instance.IActiveView
```

C#

```
ModelView IActiveView {get;}
```

C++/CLI

```
property ModelView^ IActiveView {
    ModelView^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Current active [model view](#) in this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IActiveView](#).

Example

[Access Assembly \(C++\)](#)
[Dynamic View Rotation \(C++\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::EnumModelViews Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelView::GetNext Method](#)
[IModelView::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IAddConfiguration3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddConfiguration3 Method (IModelDoc2)

Adds a new configuration to this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddConfiguration3( _
    ByVal Name As System.String, _
    ByVal Comment As System.String, _
    ByVal AlternateName As System.String, _
    ByVal Options As System.Integer _
) As Configuration
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim Name As System.String
Dim Comment As System.String
Dim AlternateName As System.String
Dim Options As System.Integer
```



```
Dim value As Configuration
```

```
value = instance.IAddConfiguration3(Name, Comment, AlternateName, Options)
```

C#

```
Configuration IAddConfiguration3(  
    System.string Name,  
    System.string Comment,  
    System.string AlternateName,  
    System.int Options  
)
```

C++/CLI

```
Configuration^ IAddConfiguration3(  
& System.String^ Name,  
& System.String^ Comment,  
& System.String^ AlternateName,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of the new configuration

Comment

Comment displayed in Configuration Properties

AlternateName

Alternate configuration name; used if swConfigOption_UseAlternateName is set to True

Options

Combination of one or more BOOLEAN configuration options as defined in [swConfigurationOptions2](#) e (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddConfiguration3](#).

Remarks

The Options argument can be a combination of any of the following values:

- swConfigOption_SuppressByDefault True if you want to suppress newly added features and mates in this configuration, false if not
- swConfigOption_HideByDefault - True if you want newly added components to be hidden, false if not
- swConfigOption_MinFeatureManager True if you want newly added components to only display their component name in the FeatureManager design tree, false if you want newly added components to display their name and each of their features in the FeatureManager design tree
- swConfigOption_DontActivate - True if you do not want the new configuration activated, false if not

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetConfigurationByName Method](#)
[IModelDoc2::GetConfigurationCount Method](#)
[IModelDoc2::GetConfigurationNames Method](#)
[IModelDoc2::IGetConfigurationByName Method](#)
[IModelDoc2::IGetCustomInfoNames2 Method](#)
[IModelDoc2::AddConfiguration3 Method](#)
[IModelDoc2::ShowConfiguration2 Method](#)
[IConfiguration Interface](#)
[IConfigurationManager Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



Adds a diameter dimension at the specified location for the selected item.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IAddDiameterDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension  
  
value = instance.IAddDiameterDimension2(X, Y, Z)
```

C#

```
DisplayDimension IAddDiameterDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ IAddDiameterDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X location for the dimension

Y
Y location for the dimension

Z
Z location for the dimension

Return Value

Newly created diameter [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddDiameterDimension2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAddDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddDimension2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::AddDimension](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IAddDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension  
  
value = instance.IAddDimension2(X, Y, Z)
```

C#

```
DisplayDimension IAddDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ IAddDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters
Y
Dimension text location in meters
Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddDimension2](#).

Remarks

You should only use this method on the visible document. Before using this method, you can check to see if the document is visible by using [ISldWorks::Visible](#).

Dimensions are created based on selection location. For example, creating an angular dimension between two lines gets different results based on which line endpoints are selected. Therefore, if you are using [IModelDocExtension::SelectById2](#) to select the entities for dimensioning, then you should specify the XYZ selection coordinates. You must also leave the object name argument empty, because if you pass the object name to IModelDocExtension::SelectById2, then the selection routines tries to locate that item without using the coordinates.

Because coordinates are ignored when the object name is passed, the dimensioning routines are not able to use a specific selection location for the dimension. This causes unpredictable results in your dimension creation because you cannot be sure which line endpoint is selected by IModelDocExtension::SelectById2.

When you use this method, you might also want to use [ISldWorks::SetUserPreferenceToggle](#) swInputDimValOnCreate to suppress the dialog box that lets the user enter the dimension value.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IAddHorizontalDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddHorizontalDimension2 Method (IModelDoc2)

Creates a horizontal dimension for the current selected entities at the specified location.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddHorizontalDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension
```

```
value = instance.IAddHorizontalDimension2(X, Y, Z)
```

C#

```
DisplayDimension IAddHorizontalDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ IAddHorizontalDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters
Y
Dimension text location in meters
Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddHorizontalDimension2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IAddOrEditConfiguration Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddOrEditConfiguration Method (IModelDoc2)

Obsolete. Superseded by [IConfiguration::GetParameters](#), [IConfiguration::IGetParameters](#), [IConfiguration::ISetParameters](#), and [IConfiguration::SetParameters](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAddOrEditConfiguration( _  
    ByVal ConfigName As System.String, _  
    ByVal ParamCount As System.Integer, _  
    ByRef ParamNames As System.String, _  
    ByRef ParamValues As System.String _  
    ) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim ConfigName As System.String  
Dim ParamCount As System.Integer  
Dim ParamNames As System.String  
Dim ParamValues As System.String  
Dim value As System.Integer  
  
value = instance.IAddOrEditConfiguration(ConfigName, ParamCount, ParamNames, ParamValues)
```

C#

```
System.int IAddOrEditConfiguration(  
    System.string ConfigName,  
    System.int ParamCount,  
    ref System.string ParamNames,
```



```
    ref System.string ParamValues
)
```

C++/CLI

```
System.int IAddOrEditConfiguration(
& System.String^ ConfigName,
& System.int ParamCount,
& System.String^% ParamNames,
& System.String^% ParamValues
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName

ParamCount

ParamNames

ParamValues

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddOrEditConfiguration](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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IAddRadialDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddRadialDimension2 Method (IModelDoc2)

Adds a radial dimension at the specified location for the selected item.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddRadialDimension2( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As DisplayDimension
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As DisplayDimension

value = instance.IAddRadialDimension2(X, Y, Z)
```

C#

```
DisplayDimension IAddRadialDimension2(
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI



```

DisplayDimension^ IAddRadialDimension2(
& System.double X,
& System.double Y,
& System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters
Y
Dimension text location in meters
Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddRadialDimension2](#).

Remarks

You can only create radial dimensions on sketch segments in an active sketch.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAddVerticalDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IAddVerticalDimension2 Method (IModelDoc2)

Creates a vertical dimension for the currently selected entities at the specified location.

.NET Syntax

Visual Basic (Declaration)

```

Function IAddVerticalDimension2( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As DisplayDimension

```

Visual Basic (Usage)

```

Dim instance As IModelDoc2
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As DisplayDimension

```



```
value = instance.IAddVerticalDimension2(X, Y, Z)
```

C#

```
DisplayDimension IAddVerticalDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ IAddVerticalDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters
Y
Dimension text location in meters
Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IAddVerticalDimension2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IClosestDistance Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IClosestDistance Method (IModelDoc2)

Calculates the distance and closest points between two geometric objects.

.NET Syntax

Visual Basic (Declaration)

```
Function IClosestDistance( _  
    ByVal Object1 As System.Object, _  
    ByVal Object2 As System.Object, _  
    ByRef Point1 As System.Double, _  
    ByRef Point2 As System.Double _  
) As System.Double
```

```

Dim instance As IModelDoc2
Dim Object1 As System.Object
Dim Object2 As System.Object
Dim Point1 As System.Double
Dim Point2 As System.Double
Dim value As System.Double

value = instance.IClosestDistance(Object1, Object2, Point1, Point2)

```

C#

```

System.double IClosestDistance(
    System.object Object1,
    System.object Object2,
    out System.double Point1,
    out System.double Point2
)

```

C++/CLI

```

System.double IClosestDistance(
    & System.Object^ Object1,
    & System.Object^ Object2,
    & [Out] System.double Point1,
    & [Out] System.double Point2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Object1

Pointer to first object

Object2

Pointer to second object

Point1

Array of x, y, z coordinates for the first point

Point2

Array of x, y, z coordinates for the second point

Return Value

Minimum distance; -1.0 if no solution

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IClosestDistance](#).

Remarks

Supported input object types include:

- swSelFACES (face)
- swSelEDGES (edge)
- swSelVERTICES (vertex)
- swSelSKETCHSEGS (sketch segments)
- swSelDATUMPLANES (reference plane)
- swSelEXTSKETCHPOINTS (point on origin)
- swSelDATUMAXES (reference axis)
- swSelCOMPONENTS (component)
- swSelREFCURVES (reference curves)

This method includes these restrictions for drawings:

- Cannot measure between a sketch entity and a model entity
- Measured sketch entities have to belong to the same sheet
- Model entity measurements are based on the model origin
- Measured object cannot be a temporary geometric entity

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateArc Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateArc Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreateArc2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreateArc( _  
    ByRef P1 As System.Double, _  
    ByRef P2 As System.Double, _  
    ByRef P3 As System.Double, _  
    ByVal Dir As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim P1 As System.Double  
Dim P2 As System.Double  
Dim P3 As System.Double  
Dim Dir As System.Short  
  
instance.ICreateArc(P1, P2, P3, Dir)
```

C#

```
void ICreateArc(  
    ref System.double P1,  
    ref System.double P2,  
    ref System.double P3,  
    System.short Dir  
)
```

C++/CLI

```
void ICreateArc(  
& System.double% P1,  
& System.double% P2,  
& System.double% P3,  
& System.short Dir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

P2

P3

Dir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateArc](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

ICreateArc2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateArc2 Method (IModelDoc2)

Creates an arc based on a center point, a start, an end point, and a direction.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateArc2( _
    ByVal XC As System.Double, _
    ByVal YC As System.Double, _
    ByVal Zc As System.Double, _
    ByVal Xp1 As System.Double, _
    ByVal Yp1 As System.Double, _
    ByVal Zp1 As System.Double, _
    ByVal Xp2 As System.Double, _
    ByVal Yp2 As System.Double, _
    ByVal Zp2 As System.Double, _
    ByVal Direction As System.Short _
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim Xp1 As System.Double
Dim Yp1 As System.Double
Dim Zp1 As System.Double
Dim Xp2 As System.Double
Dim Yp2 As System.Double
Dim Zp2 As System.Double
Dim Direction As System.Short
Dim value As SketchSegment
```

```
value = instance.ICreateArc2(XC, YC, Zc, Xp1, Yp1, Zp1, Xp2, Yp2, Zp2, Direction)
```

C#

```
SketchSegment ICreateArc2(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Xp1,
    System.double Yp1,
    System.double Zp1,
    System.double Xp2,
    System.double Yp2,
    System.double Zp2,
    System.short Direction
)
```

C++/CLI

```
SketchSegment^ ICreateArc2(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Xp1,
    & System.double Yp1,
    & System.double Zp1,
    & System.double Xp2,
    & System.double Yp2,
    & System.double Zp2,
```

& System.short [Direction](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC

X value of the circle center point in meters

YC

Y value of the circle center point in meters

Zc

Z value of the circle center point in meters

Xp1

X value of the start point of the arc in meters

Yp1

Y value of the start point of the arc in meters

Zp1

Z value of the start point of the arc in meters

Xp2

X value of the end point of the arc in meters

Yp2

Y value of the end point of the arc in meters

Zp2

Z value of the end point of the arc in meters

Direction

- +1 : Go from the start point to the end point in a counter-clockwise direction
- -1 : Go from the start point to the end point in a clockwise direction

Return Value

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateArc2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateArc2 Method](#)
[IModelDoc2::Create3PointArc Method](#)
[IModelDoc2::CreateArcByCenter Method](#)
[IModelDoc2::CreateTangentArc2 Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateCenterLine Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateCenterLine Method (IModelDoc2)

Creates a center line from P1 to P2.

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreateCenterLine( _
    ByRef P1 As System.Double, _
    ByRef P2 As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim P1 As System.Double
Dim P2 As System.Double

instance.ICreateCenterLine(P1, P2)
```

C#

```
void ICreateCenterLine(
    ref System.double P1,
    ref System.double P2
)
```

C++/CLI

```
void ICreateCenterLine(
    & System.double% P1,
    & System.double% P2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1
Array of 3 doubles (x1, y1, z1) in meters that describe the first point of the line
P2
Array of 3 doubles (x2, y2, z2) in meters that describe the second point of the line

Return Value

True if success, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateCenterLine](#).

Remarks

Use [IModelDoc2::CreateCenterLineVB](#) for Visual Basic for Applications (VBA) and other forms of Basic that do not support SafeArrays.
You can also create centerline construction geometry using [IModelDoc2::ICreateLine2](#) and [ISketchSegment::ConstructionGeometry](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCenterLine Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateCircle2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateCircle2 Method (IModelDoc2)

Creates a circle based on a center point and a point on the circle.

.NET Syntax

```
Function ICreateCircle2( _
    ByVal XC As System.Double, _
    ByVal YC As System.Double, _
    ByVal Zc As System.Double, _
    ByVal Xp As System.Double, _
    ByVal Yp As System.Double, _
    ByVal Zp As System.Double _
) As SketchSegment
```



```
Dim instance As IModelDoc2
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim Xp As System.Double
Dim Yp As System.Double
Dim Zp As System.Double
Dim value As SketchSegment

value = instance.ICreateCircle2(XC, YC, Zc, Xp, Yp, Zp)
```

```
SketchSegment ICreateCircle2(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Xp,
    System.double Yp,
    System.double Zp
)
```

```
SketchSegment^ ICreateCircle2(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Xp,
    & System.double Yp,
    & System.double Zp
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XC](#)
X value of the circle center point in meters

[YC](#)
Y value of the circle center point in meters

[Zc](#)
Z value of the circle center point in meters

[Xp](#)
X value of the point on the circle in meters

[Yp](#)
Y value of the point on the circle in meters

[Zp](#)
Z value of the point on the circle in meters

Return Value

Newly created [circle](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateCircle2](#).

Example

[Create Cylinder \(C++\)](#)

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a circle using a center point and radius, use [IModelDoc2::CreateCircleByRadius2](#) or [IModelDoc2::ICreateCircleByRadius2](#). To create a partial arc, use [IModelDoc2::CreateArc2](#) or [IModelDoc2::ICreateArc2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCircle2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateCircleByRadius Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateCircleByRadius Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreateCircleByRadius2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreateCircleByRadius( _  
    ByRef P1 As System.Double, _  
    ByVal Radius As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim P1 As System.Double  
Dim Radius As System.Double  
  
instance.ICreateCircleByRadius(P1, Radius)
```

C#

```
void ICreateCircleByRadius(  
    ref System.double P1,  
    System.double Radius  
)
```

C++/CLI

```
void ICreateCircleByRadius(  
& System.double% P1,  
& System.double Radius  
)
```



NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

Radius

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateCircleByRadius](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateCircleByRadius2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateCircleByRadius2 Method (IModelDoc2)

Creates a circle based on a center point and a specified radius.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateCircleByRadius2( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal Radius As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim XC As System.Double  
Dim YC As System.Double  
Dim Zc As System.Double  
Dim Radius As System.Double  
Dim value As SketchSegment  
  
value = instance.ICreateCircleByRadius2(XC, YC, Zc, Radius)
```

C#

```
SketchSegment ICreateCircleByRadius2(  
    System.double XC,  
    System.double YC,  
    System.double Zc,  
    System.double Radius  
)
```

C++/CLI

```
SketchSegment^ ICreateCircleByRadius2(  
& System.double XC,  
& System.double YC,  
& System.double Zc,  
& System.double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC
X value of the circle center point in meters
YC

Y value of the circle center point in meters
Zc
Z value of the circle center point in meters
Radius
Radius of the circle in meters

Return Value

Newly created [circle](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateCircleByRadius2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a circle using a center point and a point on the circle, see [IModelDoc2::CreateCircle2](#) or [IModelDoc2::ICreateCircle2](#). To create a partial arc, see [IModelDoc2::CreateArc2](#) or [IModelDoc2::ICreateArc2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateCircleByRadius2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateClippedSplines Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateClippedSplines Method (IModelDoc2)

Creates one or more sketch spline segments that are clipped against a given (x1, y1), (x2, y2) rectangle. This rectangle lies in the space of the active 2D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateClippedSplines( _  
    ByRef PropArray As System.Integer, _  
    ByRef KnotsArray As System.Double, _  
    ByRef CntrlPntCoordArray As System.Double, _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double _  
) As EnumSketchSegments
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim PropArray As System.Integer  
Dim KnotsArray As System.Double
```



```

Dim CntrlPntCoordArray As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim value As EnumSketchSegments

```

```

value = instance.ICreateClippedSplines(PropArray, KnotsArray, CntrlPntCoordArray, X1, Y1, X2, Y2)

```

C#

```

EnumSketchSegments ICreateClippedSplines(
    ref System.int PropArray,
    ref System.double KnotsArray,
    ref System.double CntrlPntCoordArray,
    System.double X1,
    System.double Y1,
    System.double X2,
    System.double Y2
)

```

C++/CLI

```

EnumSketchSegments^ ICreateClippedSplines(
    & System.int% PropArray,
    & System.double% KnotsArray,
    & System.double% CntrlPntCoordArray,
    & System.double X1,
    & System.double Y1,
    & System.double X2,
    & System.double Y2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropArray

See **Remarks**

KnotsArray

See **Remarks**

CntrlPntCoordArray

See **Remarks**

X1

x component of the lower corner of the clipping rectangle

Y1

y component of the lower corner of the clipping rectangle

X2

x component of the upper corner of the clipping rectangle

Y2

y component of the upper corner of the clipping rectangle

Return Value

Newly created [sketch segments enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateClippedSplines](#).

Remarks

The rectangle lies in the space of the active 2D sketch. The results are undefined for calls made in an active 3D sketch.

The arrays are as follows:

- *PropArray* = [*Dimension*, *Order*, *NumControlPoints*, *Periodicity*]
- *KnotsArray* = [*NumControlPoints* + *Order*]
- *CntrlPntCoordArray* = [*NumControlPoints* * *Dimension*]

If *Dimension* = 3, then *CntrlPntCoordArray* is an array of 3 doubles (x, y, z).

If *Dimension* = 4, then *CntrlPntCoordArray* is an array of 4 doubles (x, y, z, w) where w = weight.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateClippedSplines Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateEllipse Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateEllipse Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreateEllipse2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreateEllipse( _  
    ByRef Center As System.Double, _  
    ByRef Major As System.Double, _  
    ByRef Minor As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Center As System.Double  
Dim Major As System.Double  
Dim Minor As System.Double  
  
instance.ICreateEllipse(Center, Major, Minor)
```

C#

```
void ICreateEllipse(  
    ref System.double Center,  
    ref System.double Major,  
    ref System.double Minor  
)
```

C++/CLI

```
void ICreateEllipse(  
& System.double% Center,  
& System.double% Major,  
& System.double% Minor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Major

Minor

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateEllipse](#).

See Also

ICreateEllipse2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateEllipse2 Method (IModelDoc2)

Creates an ellipse using the specified center point and points.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateEllipse2( _  
    ByVal CenterX As System.Double, _  
    ByVal CenterY As System.Double, _  
    ByVal CenterZ As System.Double, _  
    ByVal MajorX As System.Double, _  
    ByVal MajorY As System.Double, _  
    ByVal MajorZ As System.Double, _  
    ByVal MinorX As System.Double, _  
    ByVal MinorY As System.Double, _  
    ByVal MinorZ As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim CenterX As System.Double  
Dim CenterY As System.Double  
Dim CenterZ As System.Double  
Dim MajorX As System.Double  
Dim MajorY As System.Double  
Dim MajorZ As System.Double  
Dim MinorX As System.Double  
Dim MinorY As System.Double  
Dim MinorZ As System.Double  
Dim value As SketchSegment
```

```
value = instance.ICreateEllipse2(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ, MinorX,  
MinorY, MinorZ)
```

C#

```
SketchSegment ICreateEllipse2(  
    System.double CenterX,  
    System.double CenterY,  
    System.double CenterZ,  
    System.double MajorX,  
    System.double MajorY,  
    System.double MajorZ,  
    System.double MinorX,  
    System.double MinorY,  
    System.double MinorZ  
)
```

C++/CLI

```
SketchSegment^ ICreateEllipse2(  
& System.double CenterX,  
& System.double CenterY,  
& System.double CenterZ,  
& System.double MajorX,  
& System.double MajorY,  
& System.double MajorZ,  
& System.double MinorX,  
& System.double MinorY,
```

& System.double [MinorZ](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX
X values for the ellipse center point
CenterY
Y values for the ellipse center point
CenterZ
Z values for the ellipse center point
MajorX
X values for a point on the ellipse and on the major axis
MajorY
Y values for a point on the ellipse and on the major axis
MajorZ
Z values for a point on the ellipse and on the major axis
MinorX
X values for a point on the ellipse and on the minor axis
MinorY
Y values for a point on the ellipse and on the minor axis
MinorZ
Z values for a point on the ellipse and on the minor axis

Return Value

Newly created [ellipse](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateEllipse2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- IModelDoc2::SetAddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call IModelDoc2::SetAddToDB(false), to restore SOLIDWORKS to its normal operating mode.
- This method also works with IModelDoc2::SetDisplayWhenAdded. If you have called IModelDoc2::SetAddToDB(True), additional performance can be gained by calling IModelDoc2::SetDisplayWhenAdded(false) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling IModelDoc2::SetDisplayWhenAdded(True).

To create a partial ellipse, use [IModelDoc2::CreateEllipticalArcByCenter](#) or [IModelDoc2::ICreateEllipticalArcByCenter](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenterVB Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateEllipticalArc2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateEllipticalArc2 Method (IModelDoc2)

Creates a partial ellipse given a center point, two points that specify the major and minor axis, and two points that define the elliptical start and end points.

.NET Syntax

Visual Basic (Declaration)

```

Function ICreateEllipticalArc2( _
    ByVal CenterX As System.Double, _
    ByVal CenterY As System.Double, _
    ByVal CenterZ As System.Double, _
    ByVal MajorX As System.Double, _
    ByVal MajorY As System.Double, _
    ByVal MajorZ As System.Double, _
    ByVal MinorX As System.Double, _
    ByVal MinorY As System.Double, _
    ByVal MinorZ As System.Double, _
    ByVal StartX As System.Double, _
    ByVal StartY As System.Double, _
    ByVal StartZ As System.Double, _
    ByVal EndX As System.Double, _
    ByVal EndY As System.Double, _
    ByVal EndZ As System.Double _
) As SketchSegment

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim CenterX As System.Double
Dim CenterY As System.Double
Dim CenterZ As System.Double
Dim MajorX As System.Double
Dim MajorY As System.Double
Dim MajorZ As System.Double
Dim MinorX As System.Double
Dim MinorY As System.Double
Dim MinorZ As System.Double
Dim StartX As System.Double
Dim StartY As System.Double
Dim StartZ As System.Double
Dim EndX As System.Double
Dim EndY As System.Double
Dim EndZ As System.Double
Dim value As SketchSegment

```

```

value = instance.ICreateEllipticalArc2(CenterX, CenterY, CenterZ, MajorX, MajorY, MajorZ, MinorX,
MinorY, MinorZ, StartX, StartY, StartZ, EndX, EndY, EndZ)

```

C#

```

SketchSegment ICreateEllipticalArc2(
    System.double CenterX,
    System.double CenterY,
    System.double CenterZ,
    System.double MajorX,
    System.double MajorY,
    System.double MajorZ,
    System.double MinorX,
    System.double MinorY,
    System.double MinorZ,
    System.double StartX,
    System.double StartY,
    System.double StartZ,
    System.double EndX,
    System.double EndY,
    System.double EndZ
)

```

C++/CLI

```

SketchSegment^ ICreateEllipticalArc2(
    & System.double CenterX,
    & System.double CenterY,
    & System.double CenterZ,
    & System.double MajorX,

```

```

& System.double MajorY,
& System.double MajorZ,
& System.double MinorX,
& System.double MinorY,
& System.double MinorZ,
& System.double StartX,
& System.double StartY,
& System.double StartZ,
& System.double EndX,
& System.double EndY,
& System.double EndZ
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CenterX

X coordinate for the ellipse center point

CenterY

Y coordinate for the ellipse center point

CenterZ

Z coordinate for the ellipse center point

MajorX

Y coordinate for a point on the ellipse and on the major axis

MajorY

Y coordinate for a point on the ellipse and on the major axis

MajorZ

Z coordinate for a point on the ellipse and on the major axis

MinorX

X coordinate for a point on the ellipse and on the minor axis

MinorY

Y coordinate for a point on the ellipse and on the minor axis

MinorZ

Z coordinate for a point on the ellipse and on the minor axis

StartX

X coordinate for counter clockwise elliptical arc start point

StartY

Y coordinate for counter clockwise elliptical arc start point

StartZ

Z coordinate for counter clockwise elliptical arc start point

EndX

X coordinate for counter clockwise elliptical arc end point

EndY

Y coordinate for counter clockwise elliptical arc end point

EndZ

Z coordinate for counter clockwise elliptical arc end point

Return Value

Newly created [ellipse](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateEllipticalArc2](#).

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#).

For COM applications, the object pointer returned from this method can be used to call any APIs on the [ISketchSegment](#) interface. The underlying [ISketchArc](#) object can be obtained using QueryInterface on the returned ISketchSegment object.

OLE applications can define a new ISketchSegment or ISketchArc object using the returned Dispatch pointer. Visual Basic applications interpret the pointer for you automatically, so you can use the returned object to call ISketchSegment or ISketchArc functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

- [IModelDoc2::SetAddToDB](#) also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call [IModelDoc2::SetAddToDB\(false\)](#), to restore SOLIDWORKS to its normal operating mode.
- This method also works with [IModelDoc2::SetDisplayWhenAdded](#). If you have called [IModelDoc2::SetAddToDB\(True\)](#), additional performance can be gained by calling [IModelDoc2::SetDisplayWhenAdded\(false\)](#) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you mu

redraw your document window (see [IModelDoc2::GraphicsRedraw2](#)) to see the entities that you added. You should also restore the original display settings by calling `IModelDoc2::SetDisplayWhenAdded(True)`.
To create a complete ellipse, use [IModelDoc2::CreateEllipse2](#) or [IModelDoc2::ICreateEllipse2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenter Method](#)
[IModelDoc2::CreateEllipticalArcByCenterVB Method](#)
[IModelDoc2::ICreateEllipticalArcByCenter Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateEllipticalArcByCenter Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateEllipticalArcByCenter Method (IModelDoc2)

Creates an elliptical arc trimmed between two points.

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreateEllipticalArcByCenter( _  
    ByRef Center As System.Double, _  
    ByRef Major As System.Double, _  
    ByRef Minor As System.Double, _  
    ByRef Start As System.Double, _  
    ByRef End As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Center As System.Double  
Dim Major As System.Double  
Dim Minor As System.Double  
Dim Start As System.Double  
Dim End As System.Double  
  
instance.ICreateEllipticalArcByCenter(Center, Major, Minor, Start, End)
```

C#

```
void ICreateEllipticalArcByCenter(  
    ref System.double Center,  
    ref System.double Major,  
    ref System.double Minor,  
    ref System.double Start,  
    ref System.double End  
)
```

C++/CLI

```
void ICreateEllipticalArcByCenter(  
& System.double% Center,  
& System.double% Major,  
& System.double% Minor,  
& System.double% Start,  
& System.double% End  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array of 3 doubles (x1, y1, z1) in meters that describe the ellipse center
Major
Array of 3 doubles (x1, y1, z1) in meters that describe a point on the ellipse and on the major axis
Minor
Array of 3 doubles (x1, y1, z1) in meters that describe a point on the ellipse and on the minor axis
Start
Array of 3 doubles (x1, y1, z1) in meters that describe the start point of the ellipse
End
Array of 3 doubles (x1, y1, z1) in meters that describe the end point of the ellipse

Return Value

True if successfully created, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateEllipticalArcByCenter](#).

Remarks

The Start and End arguments should be specified in a counter-clockwise (CCW) manner.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateEllipse2 Method](#)
[IModelDoc2::CreateEllipticalArc2 Method](#)
[IModelDoc2::CreateEllipticalArcByCenter Method](#)
[IModelDoc2::CreateEllipticalArcByCenterVB Method](#)
[IModelDoc2::ICreateEllipse2 Method](#)
[IModelDoc2::ICreateEllipticalArc2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateFeatureMgrView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateFeatureMgrView Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureMgrView( _  
    ByRef Bitmap As System.Integer _  
) As FeatMgrView
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim value As FeatMgrView  
  
value = instance.ICreateFeatureMgrView(Bitmap)
```

C#

```
FeatMgrView ICreateFeatureMgrView(  
    ref System.int Bitmap  
)
```

C++/CLI

```
FeatMgrView^ ICreateFeatureMgrView(  
& System.int% Bitmap  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateFeatureMgrView](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateFeatureMgrView2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateFeatureMgrView2 Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureMgrView2( _  
    ByRef Bitmap As System.Integer, _  
    ByVal ToolTip As System.String _  
) As FeatMgrView
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim ToolTip As System.String  
Dim value As FeatMgrView
```

```
value = instance.ICreateFeatureMgrView2(Bitmap, ToolTip)
```

C#

```
FeatMgrView ICreateFeatureMgrView2(  
    ref System.int Bitmap,  
    System.string ToolTip  
)
```

C++/CLI

```
FeatMgrView^ ICreateFeatureMgrView2(  
& System.int% Bitmap,  
& System.String^ ToolTip  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

ToolTip

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateFeatureMgrView2](#).

See Also

ICreateFeatureMgrView3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateFeatureMgrView3 Method (IModelDoc2)

Obsolete. Superseded by [IModelViewManager::CreateFeatureMgrView2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateFeatureMgrView3( _  
    ByRef Bitmap As System.Integer, _  
    ByVal ToolTip As System.String, _  
    ByVal WhichPane As System.Integer _  
) As FeatMgrView
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim ToolTip As System.String  
Dim WhichPane As System.Integer  
Dim value As FeatMgrView  
  
value = instance.ICreateFeatureMgrView3(Bitmap, ToolTip, WhichPane)
```

C#

```
FeatMgrView ICreateFeatureMgrView3(  
    ref System.int Bitmap,  
    System.string ToolTip,  
    System.int WhichPane  
)
```

C++/CLI

```
FeatMgrView^ ICreateFeatureMgrView3(  
& System.int% Bitmap,  
& System.String^ ToolTip,  
& System.int WhichPane  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

ToolTip

WhichPane

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateFeatureMgrView3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

ICreateLine Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateLine Method (IModelDoc2)

 .NET Syntax

Visual Basic (Declaration)

```
Sub ICreateLine( _  
    ByRef P1 As System.Double, _  
    ByRef P2 As System.Double _  
)
```

Visual Basic (Usage)  Copy Code

```
Dim instance As IModelDoc2  
Dim P1 As System.Double  
Dim P2 As System.Double  
  
instance.ICreateLine(P1, P2)
```

C#

```
void ICreateLine(  
    ref System.double P1,  
    ref System.double P2  
)
```

C++/CLI

```
void ICreateLine(  
& System.double% P1,  
& System.double% P2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- P1
- P2

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateLine](#).

 See Also

- [IModelDoc2 Interface](#)
- [IModelDoc2 Members](#)

ICreateLine2 Method (IModelDoc2)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreateLine2 Method (IModelDoc2)

Creates a sketch line in the currently active 2D or 3D sketch.

 .NET Syntax

Visual Basic (Declaration)

```
Function ICreateLine2( _
    ByVal P1x As System.Double, _
    ByVal P1y As System.Double, _
    ByVal P1z As System.Double, _
    ByVal P2x As System.Double, _
    ByVal P2y As System.Double, _
    ByVal P2z As System.Double _
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim P1x As System.Double
Dim P1y As System.Double
Dim P1z As System.Double
Dim P2x As System.Double
Dim P2y As System.Double
Dim P2z As System.Double
Dim value As SketchSegment
```

```
value = instance.ICreateLine2(P1x, P1y, P1z, P2x, P2y, P2z)
```

C#

```
SketchSegment ICreateLine2(
    System.double P1x,
    System.double P1y,
    System.double P1z,
    System.double P2x,
    System.double P2y,
    System.double P2z
)
```

C++/CLI

```
SketchSegment^ ICreateLine2(
    & System.double P1x,
    & System.double P1y,
    & System.double P1z,
    & System.double P2x,
    & System.double P2y,
    & System.double P2z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1x

X value of the line start point

P1y

Y value of the line start point

P1z

Z value of the line start point

P2x

X value of the line end point

P2y

Y value of the line end point

P2z

Z value of the line end point

Return Value

Newly created [ISketchSegment](#) object; if the operation fails, then NULL is returned

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreateLine2](#).

Remarks

If a sketch is not active, then the line is not created and NULL is returned. You can check for an active sketch using [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::!GetActiveSketch2](#).

For COM applications, the underlying [ISketchLine](#) object can be obtained using QueryInterface on the [ISketchSegment](#) object returned. C++ Dispatch applications can define a new ISketchLine or ISketchSegment object that uses this Dispatch pointer. Visual Basic applications interpret the pointer for you automatically so you can use the returned object to call ISketchSegment or ISketchLine functions.

[IModelDoc2::SetAddToDB](#) and [IModelDoc2::SetDisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database. IModelDoc2::SetAddToDB also avoids inferencing.

When this method is used with a drawing document, this method creates the line relative to the active drawing view, [IDrawingDoc::ActiveDrawingView](#) or [IDrawingDoc::!ActiveDrawingView](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreateLine2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePlaneAtAngle2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtAngle2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::!CreatePlaneAtAngle3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneAtAngle2( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim value As RefPlane  
  
value = instance.ICreatePlaneAtAngle2(Val, FlipDir)
```

C#

```
RefPlane ICreatePlaneAtAngle2(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneAtAngle2(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtAngle2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreatePlaneAtAngle3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtAngle3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneAtAngle3( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim AutoSize As System.Boolean  
Dim value As RefPlane  
  
value = instance.ICreatePlaneAtAngle3(Val, FlipDir, AutoSize)
```

C#

```
RefPlane ICreatePlaneAtAngle3(  
    System.double Val,  
    System.bool FlipDir,  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneAtAngle3(  
& System.double Val,  
& System.bool FlipDir,  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

Angle from the selected plane in radians

FlipDir

True to flip the plane the other way for first direction, false to not

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtAngle3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePlaneAtOffset2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtOffset2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreatePlaneAtOffset3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneAtOffset2( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneAtOffset2(Val, FlipDir)
```

C#

```
RefPlane ICreatePlaneAtOffset2(  
    System.double Val,  
    System.bool FlipDir  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneAtOffset2(  
& System.double Val,  
& System.bool FlipDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val

FlipDir

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtOffset2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreatePlaneAtOffset3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtOffset3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneAtOffset3( _  
    ByVal Val As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val As System.Double  
Dim FlipDir As System.Boolean  
Dim AutoSize As System.Boolean  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneAtOffset3(Val, FlipDir, AutoSize)
```

C#

```
RefPlane ICreatePlaneAtOffset3(  
    System.double Val,  
    System.bool FlipDir,  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneAtOffset3(  
& System.double Val,  
& System.bool FlipDir,  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val
Offset distance from selected plane in meters

FlipDir
True to flip the plane the other way for first direction, false to not
AutoSize
True to automatically size the the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtOffset3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreatePlaneAtSurface2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtSurface2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreatePlaneAtSurface3](#).

VBA Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneAtSurface2( _  
    ByVal InterIndex As System.Integer, _  
    ByVal ProjOpt As System.Boolean, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal NormalPlane As System.Boolean, _  
    ByVal Angle As System.Double _  
) As RefPlane
```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim InterIndex As System.Integer
Dim ProjOpt As System.Boolean
Dim ReverseDir As System.Boolean
Dim NormalPlane As System.Boolean
Dim Angle As System.Double
Dim value As RefPlane

value = instance.ICreatePlaneAtSurface2(InterIndex, ProjOpt, ReverseDir, NormalPlane, Angle)

```

C#

```

RefPlane ICreatePlaneAtSurface2(
    System.int InterIndex,
    System.bool ProjOpt,
    System.bool ReverseDir,
    System.bool NormalPlane,
    System.double Angle
)

```

C++/CLI

```

RefPlane^ ICreatePlaneAtSurface2(
    & System.int InterIndex,
    & System.bool ProjOpt,
    & System.bool ReverseDir,
    & System.bool NormalPlane,
    & System.double Angle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InterIndex

ProjOpt

ReverseDir

NormalPlane

Angle

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtSurface2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePlaneAtSurface3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneAtSurface3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```

Function ICreatePlaneAtSurface3( _
    ByVal InterIndex As System.Short, _
    ByVal ProjOpt As System.Boolean, _
    ByVal ReverseDir As System.Boolean, _
    ByVal NormalPlane As System.Boolean, _
    ByVal Angle As System.Double, _
)

```

```
ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim InterIndex As System.Short  
Dim ProjOpt As System.Boolean  
Dim ReverseDir As System.Boolean  
Dim NormalPlane As System.Boolean  
Dim Angle As System.Double  
Dim AutoSize As System.Boolean  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneAtSurface3(InterIndex, ProjOpt, ReverseDir, NormalPlane, Angle,  
AutoSize)
```

C#

```
RefPlane ICreatePlaneAtSurface3(  
    System.short InterIndex,  
    System.bool ProjOpt,  
    System.bool ReverseDir,  
    System.bool NormalPlane,  
    System.double Angle,  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneAtSurface3(  
& System.short InterIndex,  
& System.bool ProjOpt,  
& System.bool ReverseDir,  
& System.bool NormalPlane,  
& System.double Angle,  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InterIndex

- Multiple intersections - other solutions may exist.
- a surface, plane, and edge - the intersection index is the intersection point to use when there are multiple intersections; when the intersection index input is more than the number of intersection points, the index of the last intersection point found will be used

ProjOpt

True to project the sketch plane point along the sketch plane normal for a sketch point and a surface, false to project the sketch plane point normal to the surface

ReverseDir

True to create the plane on the opposite side of the sketch plane, false to not

NormalPlane

True to find the plane normal to the surface for a conical surface, false to find the plane tangent to the surface

Angle

Value of the angular offset of the normal plane, relative to a chosen reference plane

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneAtSurface3](#).

Remarks

This method uses the current document setting for displaying of the reference plane as it is created.

If display of reference planes is...	Then you ...
Enabled	See the reference plane on the screen as it is created
Disabled	Do not see the reference plane on the screen as it is created

The [IModelDocExtension::GetUserPreferenceToggle](#) and [IModelDocExtension::SetUserPreferenceToggle](#) methods, with swDisplayPlanes, get c set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns a RefPlane object. You can use this object for further operations on the reference plane feature. Although just having a reference plane may not be useful, it is a feature, which is an entity, so methods available on those objects are available.

For this type of user...	Those functions are...
OLE	Directly accessible
COM	Available via use of a QueryInterface

For example, if the reference plane must be selected, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ICreatePlaneFixed Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneFixed Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreatePlaneFixed2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ICreatePlaneFixed( _
    ByRef P1 As System.Double, _
    ByRef P2 As System.Double, _
    ByRef P3 As System.Double, _
    ByVal UseGlobal As System.Boolean _
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim P1 As System.Double
Dim P2 As System.Double
```



```
Dim P3 As System.Double
Dim UseGlobal As System.Boolean

instance.ICreatePlaneFixed(P1, P2, P3, UseGlobal)
```

C#

```
void ICreatePlaneFixed(
    ref System.double P1,
    ref System.double P2,
    ref System.double P3,
    System.bool UseGlobal
)
```

C++/CLI

```
void ICreatePlaneFixed(
    & System.double% P1,
    & System.double% P2,
    & System.double% P3,
    & System.bool UseGlobal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1

P2

P3

UseGlobal

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneFixed](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreatePlaneFixed2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneFixed2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager.InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneFixed2( _
    ByRef P1 As System.Double, _
    ByRef P2 As System.Double, _
    ByRef P3 As System.Double, _
    ByVal UseGlobal As System.Boolean _
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim P1 As System.Double
Dim P2 As System.Double
Dim P3 As System.Double
Dim UseGlobal As System.Boolean
Dim value As RefPlane
```

```
value = instance.ICreatePlaneFixed2(P1, P2, P3, UseGlobal)
```

C#

```
RefPlane ICreatePlaneFixed2(  
    ref System.double P1,  
    ref System.double P2,  
    ref System.double P3,  
    System.bool UseGlobal  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneFixed2(  
& System.double% P1,  
& System.double% P2,  
& System.double% P3,  
& System.bool UseGlobal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

P1
Array of 3 doubles (x, y, z) in meters; this is the first of three model-space points used to define the plane orientation; it is also used as the origin for the plane coordinate system

P2
Array of 3 doubles (x, y, z) in meters; this is the second of three model-space points used to define the plane orientation; the planes X-axis is directed from P1 to P2 unless useGlobal is set to True

P3
Array of 3 doubles (x, y, z) in meters; this is the final model-space point used to define the plane orientation

UseGlobal
Controls x-axis orientation (see **Remarks**)

Return Value

Newly created [reference plane](#) or NULL if the creation of the plan fails

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneFixed2](#).

 Remarks

The resulting plane is not parametric.

The planes normal vector is calculated using the cross product of the vectors (P2 - P1) and (P3 - P1), respectively.

The x-axis of the planes' coordinate system are a vector from P1 to P2 or a vector projected from the x-axis of the global coordinate system onto the plane.

The useGlobal argument denotes whether to align the x-axis of the plane with global orientation.

If useGlobal...	Then x-axis of the...
True	global (model) coordinate system is projected onto the plane. That vector is used to determine the direction of the plane's x-axis. This does not reorient the plane. Instead, it rotates the plane coordinate system about P1 until the x-axis of the plane aligns with the projected vector. P1, P2, and P3 are still required because they define the plane.
false	plane is aligned based on your input points, P1 and P2.

This method creates the plane in the model that is currently being edited. This behavior is consistent with the other plane creation APIs, but it is different from the original IModelDoc2::ICreatePlaneFixed, which created the plane in this model, regardless of whether a component was currently being edited.

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)

[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

Availability

SOLIDWORKS 2001Plus SP1, Revision Number 10.1

SOLIDWORKS API Help

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ICreatePlanePerCurveAndPassPoint2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlanePerCurveAndPassPoint2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreatePlanePerCurveAndPassPoint3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlanePerCurveAndPassPoint2( _  
    ByVal OrigAtCurve As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim OrigAtCurve As System.Boolean  
Dim value As RefPlane  
  
value = instance.ICreatePlanePerCurveAndPassPoint2(OrigAtCurve)
```

C#

```
RefPlane ICreatePlanePerCurveAndPassPoint2(  
    System.bool OrigAtCurve  
)
```

C++/CLI

```
RefPlane^ ICreatePlanePerCurveAndPassPoint2(  
&    System.bool OrigAtCurve  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OrigAtCurve

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlanePerCurveAndPassPoint2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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ICreatePlanePerCurveAndPassPoint3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlanePerCurveAndPassPoint3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ICreatePlanePerCurveAndPassPoint3( _  
    ByVal OrigAtCurve As System.Boolean, _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim OrigAtCurve As System.Boolean  
Dim AutoSize As System.Boolean  
Dim value As RefPlane  
  
value = instance.ICreatePlanePerCurveAndPassPoint3(OrigAtCurve, AutoSize)
```

C#

```
RefPlane ICreatePlanePerCurveAndPassPoint3(  
    System.bool OrigAtCurve,  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlanePerCurveAndPassPoint3(  
& System.bool OrigAtCurve,  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OrigAtCurve

True if you want the origin to be on the curve, false if not

AutoSize

True if plane is to be automatically sized, false if not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlanePerCurveAndPassPoint3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)

[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefPlanes Method](#)

SOLIDWORKS API Help

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ICreatePlaneThru3Points2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneThru3Points2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ICreatePlaneThru3Points3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneThru3Points2() As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneThru3Points2()
```

C#

```
RefPlane ICreatePlaneThru3Points2()
```

C++/CLI

```
RefPlane^ ICreatePlaneThru3Points2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneThru3Points2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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ICreatePlaneThru3Points3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ICreatePlaneThru3Points3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertRefPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePlaneThru3Points3( _  
    ByVal AutoSize As System.Boolean _  
) As RefPlane
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AutoSize As System.Boolean  
Dim value As RefPlane
```

```
value = instance.ICreatePlaneThru3Points3(AutoSize)
```

C#

```
RefPlane ICreatePlaneThru3Points3(  
    System.bool AutoSize  
)
```

C++/CLI

```
RefPlane^ ICreatePlaneThru3Points3(  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AutoSize

True to automatically size the plane, false to not

Return Value

Newly created [reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ICreatePlaneThru3Points3](#).

Remarks

This method uses the current document setting for displaying the reference plane. If display of reference planes is disabled, then you do not see the reference plane on the screen. If display of reference planes is enabled, then you see it. [IModelDocExtension::GetUserPreferenceToggle](#) swDisplayPlanes and [IModelDocExtension::SetUserPreferenceToggle](#) swDisplayPlanes get or set this display preference.

This method does not select the reference plane after it is created. Objects that are selected before running this method are still selected when the method completes, not the newly created reference plane.

This method returns an IRefPlane object. This object can then be used for subsequent operations on the reference plane feature. Having a IRefPlane object may not be useful, except that it is a feature, which is an entity, so methods available on those objects are available. For an OLE user, those functions are directly accessible; for a COM user, those functions are available via QueryInterface. For example, to select the reference plane, use [IEntity::Select4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::CreatePlaneAtAngle3 Method](#)
[IModelDoc2::CreatePlaneAtOffset3 Method](#)
[IModelDoc2::CreatePlaneAtSurface3 Method](#)
[IModelDoc2::CreatePlaneFixed2 Method](#)
[IModelDoc2::CreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::CreatePlaneThru3Points3 Method](#)
[IModelDoc2::CreatePlaneThruLineAndPt Method](#)
[IModelDoc2::CreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::GetVisibilityOfConstructPlanes Method](#)
[IModelDoc2::ICreatePlaneAtAngle3 Method](#)
[IModelDoc2::ICreatePlaneAtOffset3 Method](#)
[IModelDoc2::ICreatePlaneAtSurface3 Method](#)
[IModelDoc2::ICreatePlaneFixed2 Method](#)
[IModelDoc2::ICreatePlanePerCurveAndPassPoint3 Method](#)
[IModelDoc2::ICreatePlaneThru3Points3 Method](#)
[IModelDoc2::ICreatePlaneThruLineAndPt Method](#)
[IModelDoc2::ICreatePlaneThruPtParallelToPlane Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)

Availability

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